for the remaining life span of the option. This relationship is a result of the fact that greater volatility raises the probability of the intrinsic value increasing by any specified amount prior to expiration. In other words, the greater the volatility, the greater the probable price range of the market.

Although volatility is an extremely important factor in the determination of option premium values, it should be stressed that the future volatility of a market is never precisely known until after the fact. (In contrast, the time remaining until expiration and the relationship between the current market price and the strike price can be exactly specified at any juncture.) Thus, volatility must always be estimated on the basis of *historical volatility* data. The future volatility estimate implied by market prices (i.e., option premiums), which may be higher or lower than the historical volatility, is called the *implied volatility*.

On average, there is a tendency for the implied volatility of options to be higher than the subsequent *realized volatility* of the market till the options' expiration. In other words, options tend to be priced a little high. The extra premium is necessary to induce option sellers to take the open-ended risk of providing price insurance to option buyers. This situation is entirely analogous to home insurance premiums being priced at levels that provide a profit margin to insurance companies—otherwise, they would have no incentive to assume the open-ended risk.

[*]This appendix was originally published in *Market Wizards* (1989).